

BLE Log Console Quick Start Guide

Version: v1.0.5

Introduction

BLE Log Console is a terminal tool for receiving, displaying, and saving ESP BLE logs in real time. It supports UART and SPI Bridge transport modes. **Before first use, download the latest stable BLE Log Console tool package:** >[BLE Log Console tools](#)

1. Preparation

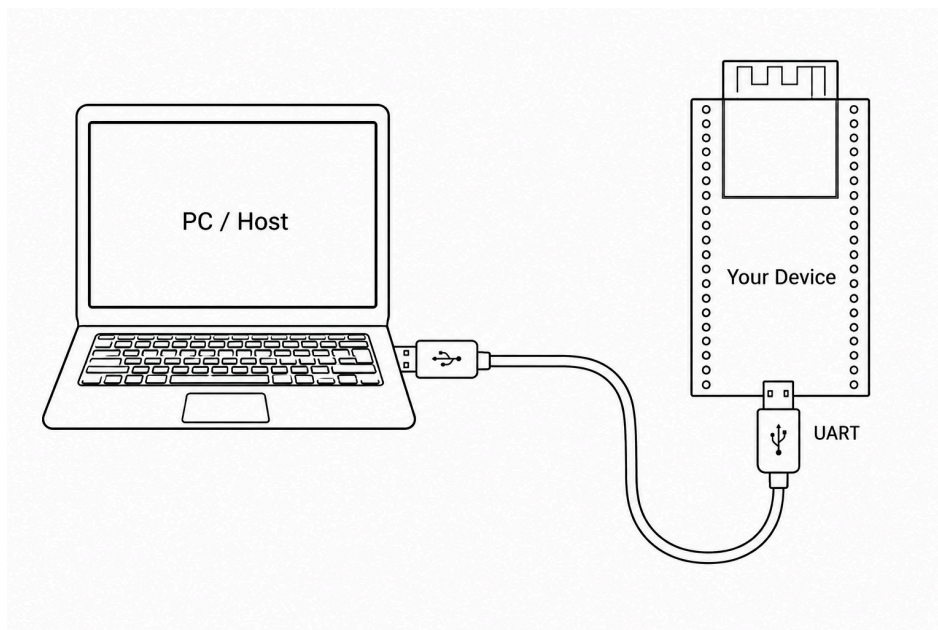
Before starting, confirm the following three items: BLE Log is enabled in the firmware, the hardware is wired correctly, and the PC-side tool can run.

Jump directly to the transport mode you want to use:

- [UART Mode](#)
- [SPI Bridge Mode](#)

2. UART Mode

Device connection diagram:



2.1 Firmware Configuration

- Enable `CONFIG_BT_LOG_CRITICAL_ONLY` in the existing project, and configure the related options as follows:
 - `CONFIG_BLE_LOG_PRPH_UART_DMA=y`
 - `CONFIG_BLE_LOG_PRPH_UART_DMA_PORT=0`
 - `CONFIG_BLE_LOG_PRPH_UART_DMA_BAUD_RATE=3000000`
 - `CONFIG_BLE_LOG_PRPH_UART_DMA_TX_IO_NUM=0`
- Confirm that UART TX is connected to the PC-side serial port RX.
- Build and flash the firmware.

Notes: - CONFIG_BLE_LOG_PRPH_UART_DMA_TX_IO_NUM sets the GPIO corresponding to the **UART TX signal**. The default value is 0. - If UART0 is already occupied by another function, adjust CONFIG_BLE_LOG_PRPH_UART_DMA_PORT according to the actual hardware connection to avoid affecting existing functionality. - CONFIG_BLE_LOG_PRPH_UART_DMA_BAUD_RATE is the baud rate used by UART to output BLE logs. The baud rate selected in the PC-side ble_log_console must match this value.

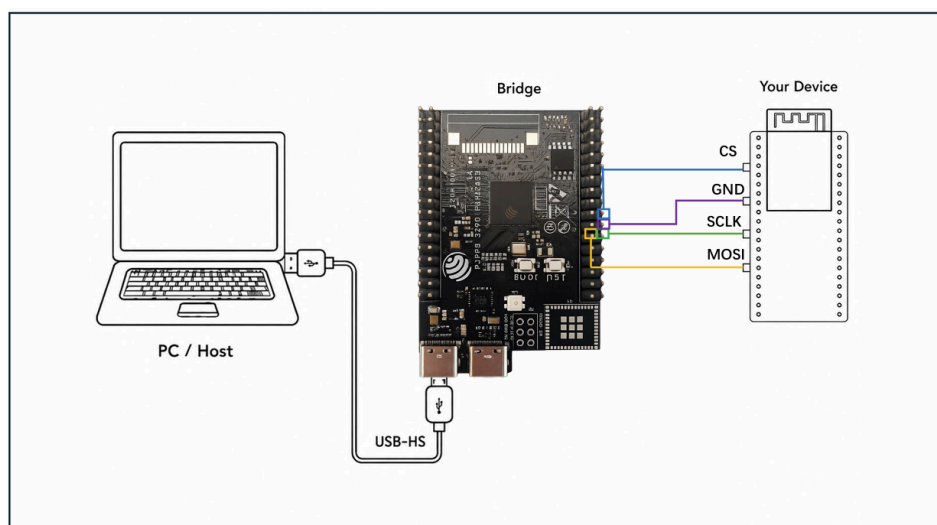
2.2 Start the Application on the PC

- For Windows instructions, see [Windows System](#).
- For Linux instructions, see [Linux System](#).

3. SPI Bridge Mode

Device connection diagram:

ESP device → BLE Log SPI USB Bridge → PC



How do I get a Bridge device?

The Bridge device forwards **BLE SPI Log data generated by your device** to the PC. It is currently supported by our ESP32P4 product. You can get a Bridge device in either of the following ways:

- Flash it yourself: If you have an ESP32P4 development board, connect it through a USB serial port and flash the firmware on the [Bridge firmware flashing](#) page. After flashing, it can be used as a Bridge device.
- Apply for or purchase one officially: If you do not have an ESP32P4 device that can be used as a Bridge, but you do need SPI Bridge mode, you can purchase one through official channels or apply for a trial device through technical support.

Notes: * The Bridge device is an additional adapter device. It cannot be the same device that generates BLE Log data. * If the flashing method above does not work smoothly, you can manually download the firmware from [Bridge firmware manual download](#) and flash it yourself.

3.1 Firmware Configuration

- Enable CONFIG_BLE_LOG_ENABLED in the target device project that generates BLE Log, and also enable the following related option:
 - CONFIG_BLE_LOG_PRPH_SPI_MASTER_DMA=y
- Configure the following GPIO numbers according to the actual circuit:
 - CONFIG_BLE_LOG_PRPH_SPI_MASTER_DMA_MOSI_IO_NUM

- CONFIG_BLE_LOG_PRPH_SPI_MASTER_DMA_SCLK_IO_NUM
- CONFIG_BLE_LOG_PRPH_SPI_MASTER_DMA_CS_IO_NUM
- Connect the corresponding GPIOs to the SPI Bridge:
 - MOSI_IO_NUM to Bridge IO 4
 - SCLK_IO_NUM to Bridge IO 5
 - DMA_CS_IO_NUM to Bridge IO 6
 - Connect the GND pins of both devices
- Build and flash the firmware.

Notes: - **The GPIO configuration above must match the actual wiring.** - If the GND pins of the two devices are not connected correctly, data reception may be abnormal. - CONFIG_BLE_LOG_ENABLED is the global BLE Log switch. If it is not enabled, the PC side will not receive BLE log data.

3.2 Start the Application on the PC

- For Windows instructions, see [Windows System](#).
- For Linux instructions, see [Linux System](#).

4. First Run and Startup

4.1 Get the Tool

The tool package provides applications for different operating systems. Download the latest tool package from the [BLE Log Console stable release](#) page.

4.1.1 Windows System

On Windows, use ble_log_console_windows_v1.0.5.exe. The recommended way to start it is to double-click the icon.

Note: This tool package currently supports Windows 10 and later.

4.1.2 Linux System

On Linux, use ble_log_console_ubuntu_v1.0.5. If the file does not have execute permission, enter the corresponding directory first and run:

```
chmod +x ./ble_log_console_ubuntu_v1.0.5
```

If you plan to use SPI Bridge on Linux, run the following command before the first use:

```
sudo ./ble_log_console_ubuntu_v1.0.5
```

This command installs the USB access permission rules required by SPI Bridge. After the command finishes, unplug and reconnect the SPI Bridge device once. After that, sudo is no longer required for normal use.

Then run the following command to start the program:

```
./ble_log_console_ubuntu_v1.0.5
```

Note: This tool package currently supports Ubuntu 22.04 and later.

4.2 Application Guide

After startup, the application displays the following screen:

BLE Log Console Setup

Language / 语言

简体中文 ▼

Transport Mode

UART ▼

Port

/dev/ttyUSB0 ▼ Refresh

Baud Rate

3000000 ▼

Log Directory

/home/guozifan/workspace/ble_t Browse...

First, choose the language used for warnings and the quality report, then select the transport mode. Select UART if you receive logs through a serial port, or SPI Bridge if you receive logs through an SPI Bridge device. After the mode is selected, the application automatically scans the available ports.

Then select the port. In UART mode, you also need to select the baud rate, which must match the firmware configuration. Next, specify the log save path. By default, logs are saved to the `logs` directory under the current directory; if this directory does not exist, the application creates it automatically. Finally, click **Connect** to start receiving logs.

To end a capture, click **Stop & Review**, or press `q` / `Ctrl+C`. The tool first stops reception and saves remaining data, then waits up to 20 seconds for the quality check before displaying the report. It does not exit immediately. From the report, you can record again with the same configuration or exit. Large captures may be split into multiple `part` files.

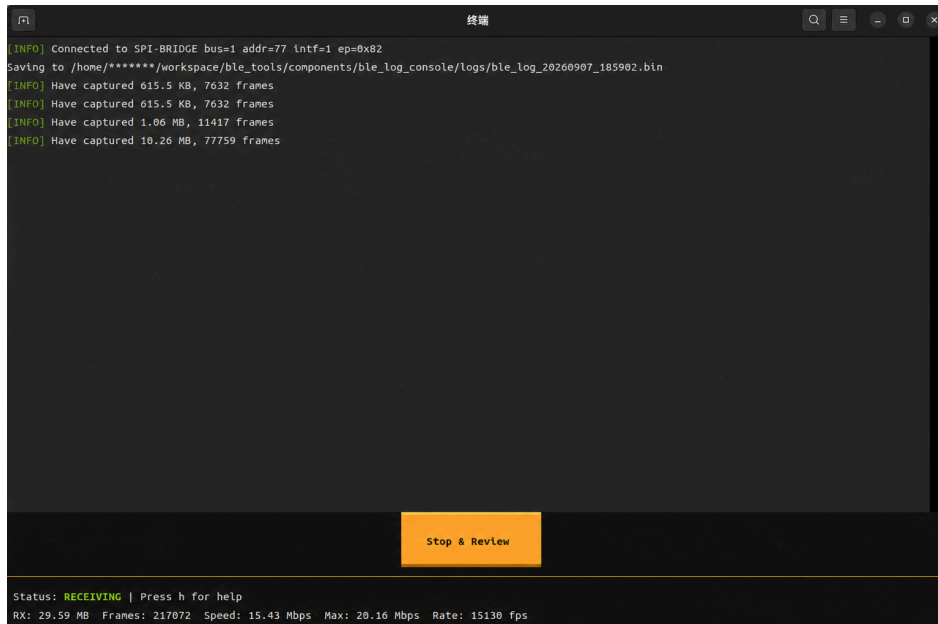
Note: End capture with **Stop & Review**, `q`, or `Ctrl+C`. While `FINALIZING` is displayed, wait for the tool to finish saving. Closing the terminal window directly may leave the final data and quality report incomplete.

On Linux, UART and SPI Bridge may show different port name formats. Select a device from the current mode's list.

5. User Interface and Log Files

5.1 Interface Description

After startup, the interface mainly consists of the log area and the status bar, as shown below.



The log area displays forwarded serial logs, status messages, and warnings in real time.

The status bar shows the connection state, amount of data received by the PC, current speed, peak speed, and frame rate. Use the quality report after recording to check for data problems.

The tool supports adaptive window resizing. When the window is small, some status information may not fit on screen. Enlarge the window or scroll to view the remaining information.

5.2 Normal Status Description

As shown above, `Frames` is the number of valid log frames recognized by the tool. `RX` is the amount of data received and saved by the PC.

Normal status: If both `RX` and `Frames` keep increasing, the tool is receiving and recognizing logs normally. If `RX` increases but `Frames` does not increase for a long time, first check the transport mode, UART baud rate, firmware configuration, and wiring.

If the connection is normal but the indicators above are abnormal, check whether the BLE Log module is enabled properly and whether the wiring is fully correct.

Note: During high traffic, the screen may update more slowly, but the tool still gives priority to saving data. If saving fails, the screen shows an explicit error.

5.3 Log File Save Rules

Recording files are saved to the log directory selected at startup. If the save path is not changed, they are saved to the `logs` directory under the current directory. File names are generated from the start time:

```
ble_log_YYYYMMDD_HHMMSS.bin
```

If serial-log forwarding is enabled in the firmware, regular console logs that would normally appear on UART0 (the serial monitor) are also saved as:

```
ble_log_YYYYMMDD_HHMMSS_console.log
```

After a safe stop, the matching quality report is saved as:

```
ble_log_YYYYMMDD_HHMMSS_report.txt
```

After the program exits, it prints the actual save paths in the terminal. Large recordings are automatically split into multiple files. If another recording starts within the same second, a numeric suffix is added so existing files are not overwritten.

5.4 Capture Quality Report

After recording stops, the tool shows one of three conclusions:

- **READY FOR ANALYSIS:** The data was saved successfully and can be decoded. It is ready to submit for analysis.
- **SAVED WITH WARNINGS:** The data was saved, but the recording may contain an interruption, missing data, or a condition that could not be verified. The files can still be submitted to our technical support.
- **RECAPTURE RECOMMENDED:** The recording did not contain enough usable data, the data could not be saved correctly, or the detected loss rate was high. Check the connection, mode, port, baud rate, and firmware configuration, then record again. If this happens repeatedly, contact technical support for assessment.

The report screen shows the captured frame count, data size, and file locations. For further troubleshooting, provide both the generated `report.txt` and `.bin` files to technical support.

6. Further Information

Press `h` for more shortcuts. For source setup and command-line options, see the [repository README](#).

Appendix: Troubleshooting and Notes

Quick links:

- [Device not shown in the port list](#)
- [Program starts but no logs appear](#)
- [Serial-monitor log display and saving](#)
- [Source environment setup fails](#)
- [Confirming ESP32P4 Bridge firmware version](#)
- [Quality report is unverified or the check is incomplete](#)

I plugged in the device, but I still cannot see it in the port list.

Please check:

- Whether the device is wired correctly, including whether GND is properly shared in SPI mode.
- Whether the device is powered on.
- Whether the correct firmware is running on the device.
- Whether the port is already occupied by another program.
- On Linux, whether you have run `sudo ./ble_log_console_ubuntu_v1.0.5` once and unplugged/reconnected the device after running it.

The program starts, but no logs appear.

Please check:

- Whether the selected connection mode is correct.
- Whether the selected port is correct.
- Whether BLE Log is enabled in the ESP device firmware.
- In UART mode, whether the baud rate selected in the tool matches the firmware configuration.
- Whether the wiring between the ESP device and the serial port tool or SPI Bridge is correct.

- In SPI Bridge mode, whether the GPIOs corresponding to MOSI, SCLK, and CS in the ESP device firmware match the actual wiring.

Why do serial-monitor logs appear in the tool, and where are they saved?

When serial-log forwarding is enabled in the firmware, regular console logs that would normally appear on UART0 (the serial monitor) are forwarded to BLE Log Console together with the BLE Log data. The tool displays them in the log area and automatically saves them to `ble_log_YYYYMMDD_HHMMSS_console.log` in the selected log directory.

The first line of each console-log group shows when the PC received it, for example `[15:14:54.367]`. Following lines without a timestamp belong to the same group. This is not the device runtime.

The `_console.log` file automatically removes color and other display-control characters and normalizes line endings. The original `.bin` file is unchanged.

Source environment setup fails.

Source startup depends on the local Python environment, package index access, and system PATH settings. The install scripts try to prepare these automatically, but different PC environments may need manual setup. The required items are:

- Python 3.11 or later.
- `uv`, available from PATH.
- Access to the Python package dependencies declared in `pyproject.toml`.

If the script cannot complete the setup, install the required items manually, then run `uv sync --all-extras` in the source directory before using the startup script.

How do I confirm the firmware version flashed on ESP32P4?

The current ESP32P4 Bridge firmware version is 1.0. The firmware prints version information in the serial log when it starts. To confirm the version, connect the ESP32P4 to the PC through the USB serial port, then open the serial monitor and check the startup log.

```
idf.py -p <PORT> monitor
```

If you missed the startup log, press the reset button on the development board. The version information is printed again after ESP32P4 restarts.

Note: When checking the version, the ESP32P4 serial port cannot be used by `ble_log_console` in UART mode or another serial tool at the same time.

Can I use the files if the quality report is unverified or the check is incomplete?

The report separately states whether the raw data was saved completely and how much data the automated quality check covered. If the check does not finish within 20 seconds, the raw files are still retained. Frame and loss results then cover only the parsed portion, and continuity for the full recording is shown as unverified.

The detailed report lists received frames, firmware write failures, and sequence checks for each segment, and marks segments with an incomplete start or end. When firmware segment statistics are available, overall quality statistics use only complete segments. If continuity cannot be determined, it is shown as unverified.

Keep the original `.bin` files and matching `_report.txt`. If the recording was split, provide all parts so technical support can assess whether another capture is needed.